Week 2
Data Cleaning & Quality Report

What This Report Covers

Week 1 identified the patterns in the data. Week 2 fixed the problems that were blocking reliable analysis. This report documents every cleaning step taken, shows what changed, and confirms which analyses are now ready to run.

The dataset started with 7,543 records across 80 columns. After cleaning, it contains 5,000 unique records across 44 usable columns. All institution names, applicant names, contact details, and identifying information have been removed from this public version.

7,543 → 5,000 **Records (raw → clean)** 2,543 duplicates removed	**80 → 44** **Columns (raw → clean)** 36 empty/redundant dropped	**9** **Data type errors fixed** across 9 columns	**7** **Cleaning steps applied** in sequence

Section 1: What Was Cleaned and Why

Seven cleaning steps were applied in sequence. Each step targeted a specific problem that would have produced wrong results if left unfixed.

Step	What Was Done	Problem It Fixed	Why It Matters
1	**Removed 38 completely empty columns**	38 of 80 columns had zero values	Empty columns inflate the schema and trigger false alarms in any quality check. No real data was lost.
2	**Fixed 3 misspelled column names**	Column headers had typos (e.g. Recieved_At, Counsler)	Misspelled names break every script that references those fields by name. Fixing at source stops the error spreading.
3	**Removed 2,543 duplicate rows**	The same application appeared up to 9 times in the data	Duplicates inflated all metrics by 33.7%. One program showed 1,610 raw applications against a true count of 1,080.
4	**Standardised country name formatting**	Mixed casing split one country into multiple values (e.g. 'India' and 'india')	Any count, map, or filter by country would treat these as different places and produce wrong totals.
5	**Converted 7 date columns from numbers to dates**	Dates were stored as 13-digit numbers (milliseconds)	Number-format dates cannot be filtered by date range or used in any time-based chart or calculation.
6	**Split the combined Intake field into two columns**	Program name and intake term were bundled in one field	Separating them into Program and Term columns allows independent filtering on either dimension.
7	**Fixed phone number data type**	176 phone numbers were stored as integers	Integer storage drops leading zeros and allows invalid arithmetic on what should be a text identifier.

Data Type Corrections (Second Pass)

After the initial seven steps, a full audit of every column's data type identified nine additional corrections. These were applied in a second pass.

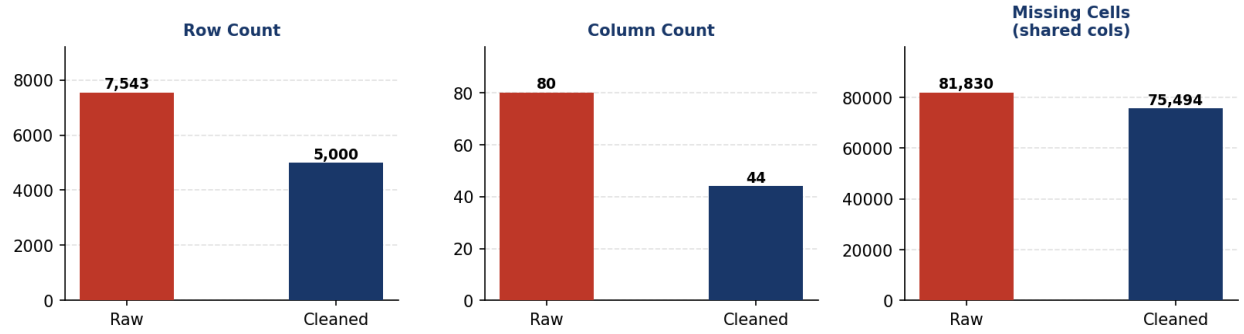
Column	Before	Correction and Reason

Reference ID	Stored as a number	Changed to text. Prevents invalid arithmetic on an ID field.
Reference ID-2	Stored as a decimal (float)	Changed to text. Removed the .0 artifact (e.g. 291925.0 → 291925).
Campaign ID	Stored as a decimal	Changed to text. Campaign codes are labels, not quantities.
Record ID	Stored as a decimal	Changed to text. Same reason as Campaign ID.
Attempts	Stored as a decimal	Changed to a whole number. Outreach attempt counts should not have decimal places.
Admit Date	Mixed text and date values	Unified to a single date format.
Received At 2	Stored as text strings	Converted to proper date format (was stored as '04/29/2024 00:00:00').
Date of Contact	Stored as text strings	Converted to proper date format, consistently with Received At 2.
Postal Code	Stored as a number	Changed to text. Prevents leading zeros being dropped.

Section 2: Before and After — The Charts

Chart 1: Row Count, Column Count, and Missing Cells

This chart compares three core measures of the dataset before and after cleaning. Red bars show the raw data. Dark blue bars show the cleaned data.

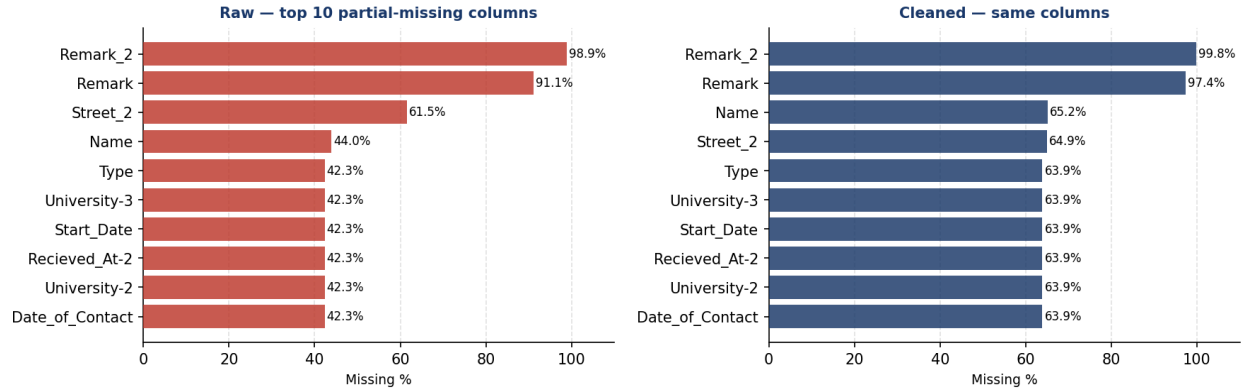


What This Shows

Row count dropped 33.7% after duplicates were removed. Column count halved after 38 empty columns were dropped. Missing cells in the remaining shared columns fell from 81,830 to 75,494.

Chart 2: Missing Data — Before and After Cleaning

These two charts show the top 10 columns with partial missing data. The left panel shows the raw data. The right panel shows the same columns after cleaning.

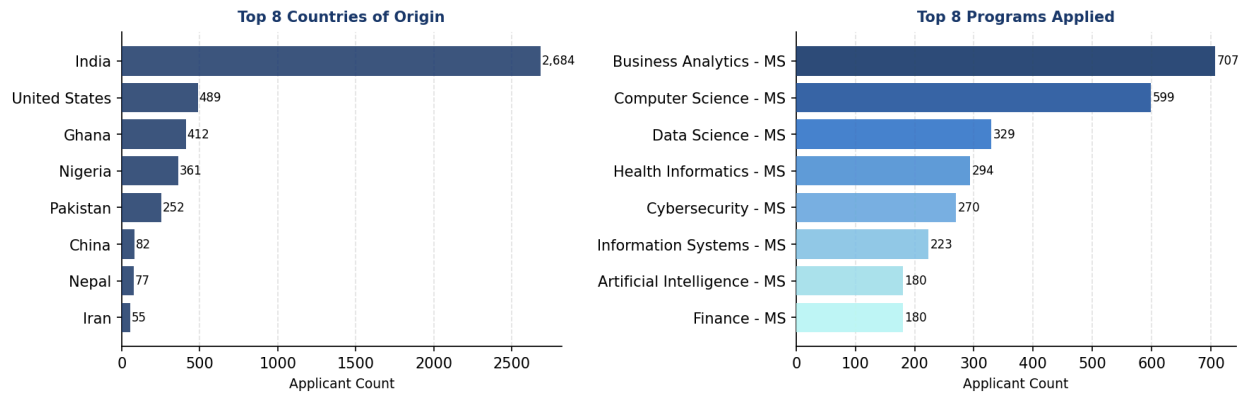


What This Shows

The Admit Date column improved from 23.3% missing to 13.9% missing, largely because many duplicate rows were also missing this value and were removed. The Remark columns remain high-null by design — they are optional free-text notes that counselors fill in selectively.

Chart 3: Who Is Applying and For What

These two charts show the top 8 countries of origin and the top 8 programs in the cleaned 5,000-record dataset.

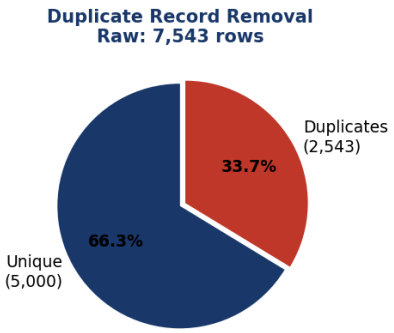


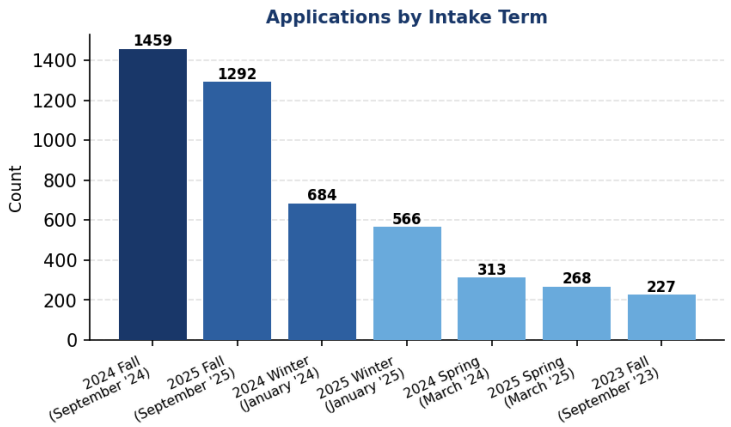
What This Shows

One country accounts for 2,684 applicants — 53.7% of the cleaned dataset. Business Analytics MS (707) and Computer Science MS (599) together represent 35.8% of all applications. These figures are reliable post-cleaning, unlike the inflated Week 1 raw counts.

Chart 4: Duplicate Removal and Application Volume by Term

The left chart shows what proportion of the raw data was duplicates. The right chart shows application volume by intake term after cleaning.





What This Shows

One in three raw records was a duplicate. After removal, 2024 Fall is the largest cohort at 1,459 applications, followed by 2025 Fall at 1,292. The term-level breakdown is now accurate and reliable for counselor workload planning.

Section 3: Full Before vs After Comparison

The table below documents every measurable change between the raw dataset and the cleaned version.

Measure	Raw Data	Cleaned Data	Change
Total rows	7,543	5,000	-2,543 (-33.7%)
Total columns	80	44	-36 (-45.0%)
Completely empty columns	38	0	All removed
Duplicate applicant rows	2,543	0	All removed
Column name typos	3	0	Corrected
Incorrect data types	9	0	Fixed in second pass
Date columns stored as numbers	7	0	All converted to dates
Combined fields (unparsed)	1	0	Split into Program + Term
Country name inconsistencies	Mixed	Title Case	Standardised
Phone numbers stored as integers	176	0	All converted to text
Missing — Admit Date	23.3%	13.9%	-9.4 percentage points
Missing — Phone Number	7.8%	8.3%	Slight rise (denominator effect*)
New columns added	0	2	Program and Term (split from Intake)

** Phone number missing rate rose slightly because duplicate rows that were also null were removed, shrinking the denominator faster than the numerator.*

Section 4: What Each Problem Would Have Caused

Every problem in the raw data had a specific consequence. The table below shows what each issue would have produced in any analysis built before cleaning was done.

Problem	If Left Uncleaned	Fixed?
2,543 duplicate rows	Business Analytics MS count would show 1,610 instead of the true 1,080. Every program metric overstated by 33.7%.	**Yes**
38 empty columns	Quality tools would flag 38 critical missing-data fields. The schema would appear 80 columns wide when only 44 contain any data.	**Yes**
9 incorrect data types	Reference IDs sortable as numbers. Date filters would throw errors. Postal codes would drop leading zeros.	**Yes**
Combined Intake field	Filtering by program or by term would require manual text parsing on every single query.	**Yes**
Dates stored as numbers	All date range filters and time-series charts would fail with type errors.	**Yes**
Mixed country name casing	'India' and 'india' counted as two separate countries. Geographic maps and filters would be unreliable.	**Yes**

Section 5: Known Limitations and What Is Still Missing

Cleaning resolved all six structural problems. However, four data gaps remain that require a re-export from the source system. These are not fixable through cleaning alone.

Status Column — 100% Empty

The admissions status field contains no data. Conversion rate analysis and the full admissions funnel cannot be built until this is re-exported from the source system.

Counselor Column — 100% Empty

The field that identifies which counselor handled each application is entirely blank. Workload analysis and performance tracking require a re-export.

Admit Date — 13.9% Missing

Conversion rate calculations apply to 86.1% of records only. Decision timeline analysis is incomplete until the remaining dates are recovered.

Region — 14.0% Missing

Sub-national geographic analysis is incomplete. Country-level analysis is unaffected (only 0.2% null), but anything below country level should be treated as partial.

College vs First Choice College

These two fields differ in nearly every row. Their exact meaning needs clarification from the source system before either can be used in analysis.

Section 6: What Is Ready for Week 3

The table below shows which analyses can proceed immediately, which cannot, and why.

Analysis	Ready?	Detail
Application volume by program or term	**Yes**	Program and Term columns are complete and correctly typed
Country-level geographic analysis	**Yes**	Country names standardised, only 0.2% null
Date range filtering and time-series charts	**Yes**	All 7 date columns converted to proper date format
Admissions funnel and conversion rate	**No**	Status column is 100% empty — re-export from source needed
Counselor workload analysis	**No**	Counselor column is 100% empty — re-export from source needed
Sub-national geographic breakdown	**Partial**	Region column is 14% null — country level is fine

Conclusion

The dataset is now clean and structured for analysis. All six structural problems from Week 1 have been resolved. Three analyses can start immediately: program volume, geographic breakdown, and time-series filtering. Two analyses (admissions funnel and counselor workload) require a data re-export before they can proceed. Week 3 will focus on building dashboards and running the analyses that are ready.

Skills and Tools Used

Tool / Skill	How It Was Applied
Python — pandas	Deduplication, data type corrections, column standardisation, descriptive statistics
Python — matplotlib	All four charts in this report
Exploratory Data Analysis	Guided identification of quality issues in Week 1 and verification post-cleaning in Week 2
Data Quality Assessment	Systematic classification and resolution of six categories of structural problems

Tidy Data principles	Informed the decision to split the combined Intake field into separate Program and Term columns
ISO standards reference	Country name normalisation (ISO 3166) and phone format standardisation (E.164)